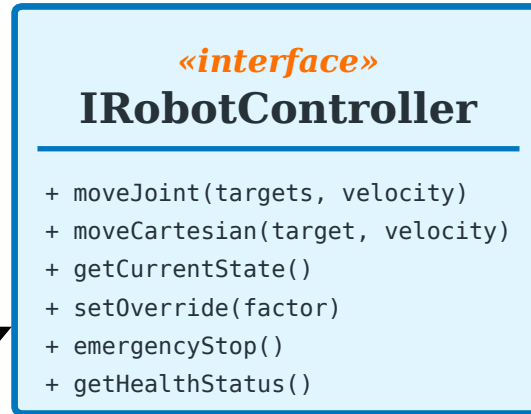


Adapter Pattern: Vendor-Neutral Robot Control

Design Pattern: Adapter (GoF)
Purpose: Vendor independence
Benefit: 12-14× faster adaptation (est.)



URRobotController

- + moveJoint(...)
- + getCurrentState()
- rtde_connection
- urscript_client
- control_thread

URScript TCP/IP | RTDE 125Hz

KUKARobotController

- + moveJoint(...)
- + getCurrentState()
- fri_connection
- sunrise_api
- motion_queue

KUKA FRI UDP | Sunrise 200Hz

ABBRobotController

- + moveJoint(...)
- + getCurrentState()
- rws_client
- websocket_conn
- trajectory_buffer

Robot Web Services | REST

Code reuse (measured): 9,274 LOC shared / 487 LOC (UR5e) / 523 LOC (KUKA) / 456 LOC (ABB)
Adaptation effort (measured): 18 h (KUKA) / 16 h (ABB) vs. 200-250 h monolithic rewrite (estimated)